🎬 Data Analysis Report: How to Make 1000% ROI in Movie Production

Prepared for Fledgling Movie Studio

📌 Key Findings (Executive Impact Summary)

This analysis identifies a repeatable production strategy capable of generating exceptional returns when specific financial and creative constraints are applied.

Analysis of more than **7,000 movies released since 1990** reveals a clear pattern capable of **producing, approaching or exceeding 1,000% Return on Investment (ROI)** under optimal conditions.

The most important findings are:

• **Horror films with budgets of $10 million or less generate an average portfolio ROI of 872%**, making them the most profitable genre–budget combination in the dataset.

• Within this category:
• **PG-13 rated films outperform R-rated titles (1,197% vs. 759% ROI)**
• **90–120-minute runtimes outperform shorter films (980% vs. 688% ROI)**
• **October, April, January, and February are the strongest release months**, with October producing the highest overall returns

• When multiple optimal characteristics align, historical ROI **exceeds 1,000%**, far surpassing the industry-wide average of **187%**.

• Writer Leigh Whannell and director James Wan demonstrate consistently higher ROIs than others in their field, with their collaborations averaging approximately **4,600% ROI**.

Together, these findings provide a clear blueprint for deploying capital efficiently while minimizing downside risk.

🎯 Problem Statement

This analysis was conducted to answer a single strategic question: how can a new studio maximize ROI while managing financial risk?

Fledgling Movie Studio is standing on the precipice of a significant financial opportunity. Armed with **$100 million in venture capital**, the studio aims to produce a movie—or slate of movies—with the **highest possible Return on Investment (ROI)**.

By maximizing ROI, the studio seeks to secure future investment, attract strategic partners, and establish itself as a credible competitor in the film industry.

This report identifies which production and release decisions historically lead to the strongest financial outcomes and translates those insights into actionable recommendations.

📊 Executive Summary

Rather than evaluating films individually, this analysis treats movie production as a **portfolio investment problem**.

A **portfolio-based ROI approach** was used, aggregating total budgets and total gross revenue within each category before calculating ROI.

Using this approach:

• The **industry-wide ROI** since 1990 is **187%**

• **Low-budget horror films (≤ $10M)** outperform every other category with **872% ROI**

• Additional optimization through rating, runtime, release month, and creative talent can push returns well beyond this baseline.

🗂️ Data Sources

The dataset used for this analysis was sourced from Kaggle and contains over **7,000 movies scraped from IMDb**:

https://www.kaggle.com/datasets/danielgrijalvas/movies

The data was cleaned, standardized, and prepared using **Excel and SQL**, and the analysis was conducted using **SQL Server and Power BI**.

⚠️ Data Limitations

Several limitations should be acknowledged:

• **Genre classification:** Only the first listed IMDb genre was used.

• **Small-sample categories:** Some analyses include categories with relatively small sample sizes.

• **Extreme outliers:**
The Blair Witch Project (≈ **414,299% ROI**) and
Paranormal Activity (≈ **1,288,939% ROI**) were excluded because they would distort the analysis.

• **ROI interpretation:** ROI values represent **group-level portfolio returns**, not individual movie averages, so as not to bias the analysis towards small budget outliers.

🛠️ Analysis Steps

1. **Data Extraction**
   Retrieved movie data from Kaggle's IMDb dataset.
2. **Data Cleaning and Formatting**
   Cleaned and standardized the data using **Excel and SQL**.
3. **Data Transfer to SQL**
   Loaded the data into **SQL Server** for analysis.
4. **Data Transformation and Modeling**
   Calculated ROI and built analytic views for **Power BI visualization**.

📈 Findings

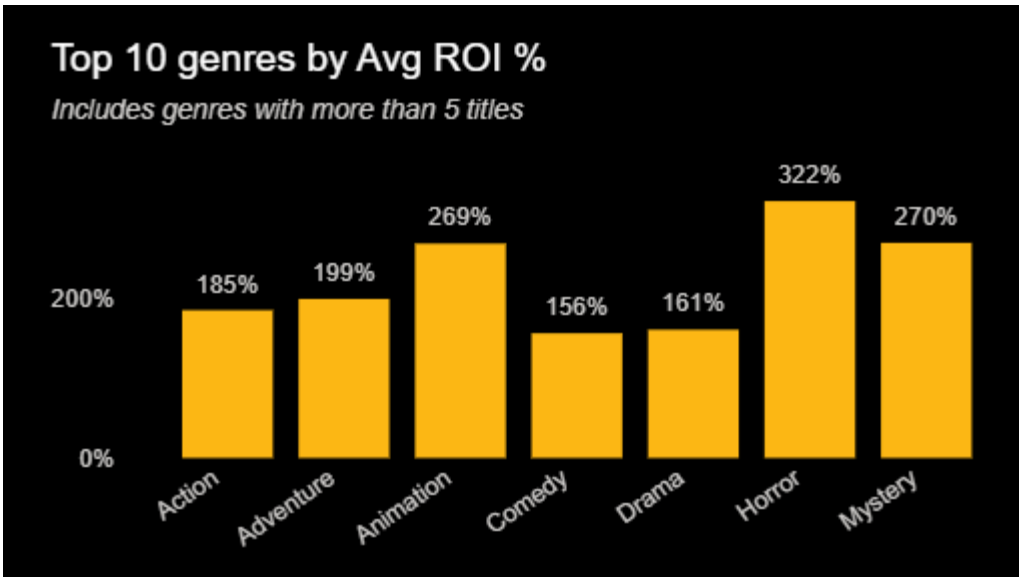
1. Genre-Level ROI and Budget Impact

To ensure meaningful cross-genre comparisons, this analysis includes only genres with **six or more films** when evaluating ROI at the genre level.

Across all budgets:

- **Horror averages a portfolio ROI of 322%**
- **Mystery averages 270%**

This indicates that Horror performs better than all other genres at the aggregate level.

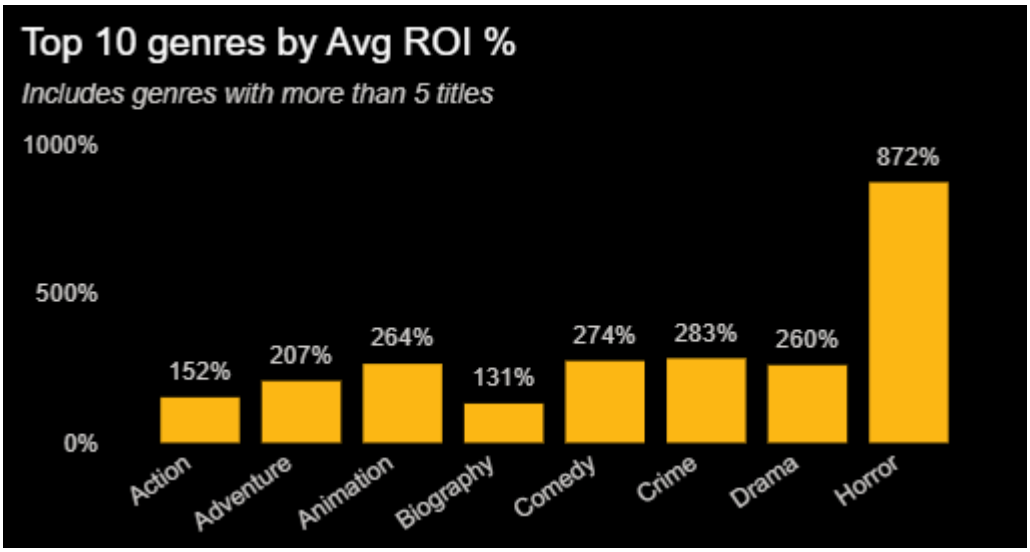


While Horror performs better than all other genres with a **322% ROI (over 50 percentage points higher than the next highest genre)**, its true advantage becomes clear when budgets are constrained…

💰 Budget Impact (≤ $10M)

When movie budgets are limited to **$10 million or less**, Horror becomes a clear outlier:

Genre	ROI
Horror	872%
Crime	283%
Comedy	274%



Conclusion

While Horror performs well across all budgets, **low-budget horror films dramatically outperform every other genre in the same financial category**, delivering more than **three times the ROI** of the next highest genre.

🔎 ANALYSIS SCOPE FOR ALL REMAINING FINDINGS

All remaining findings focus **exclusively** on:

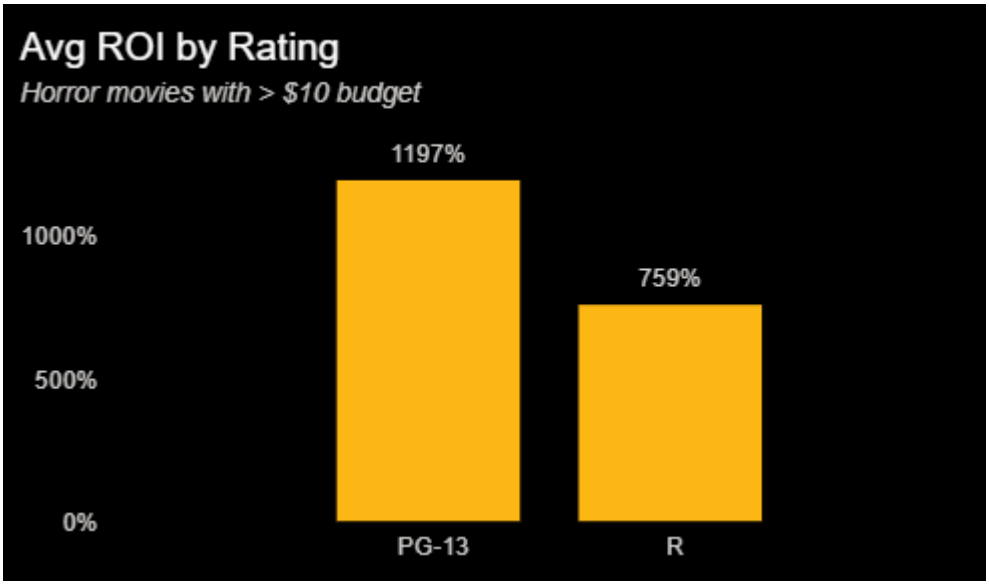
Horror movies with budgets ≤ $10 million (Base ROI = 872%)

⭐ 2. ROI by Rating

Audience accessibility plays a meaningful role in financial performance.

Rating	ROI
PG-13	1,197%
R	759%

Rating	ROI
PG-13 + R	898%

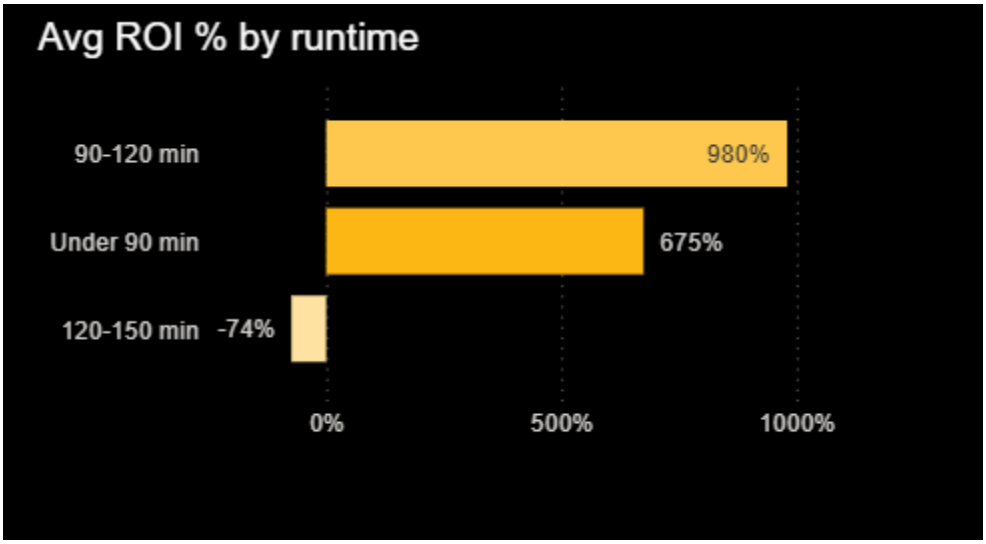


PG-13 horror films significantly outperform R-rated titles, which are still well above the industry average.

⏱️ 3. Runtime Considerations

Runtime length influences financial performance.

Runtime	ROI
90–120 minutes	980%
Under 90 minutes	688%

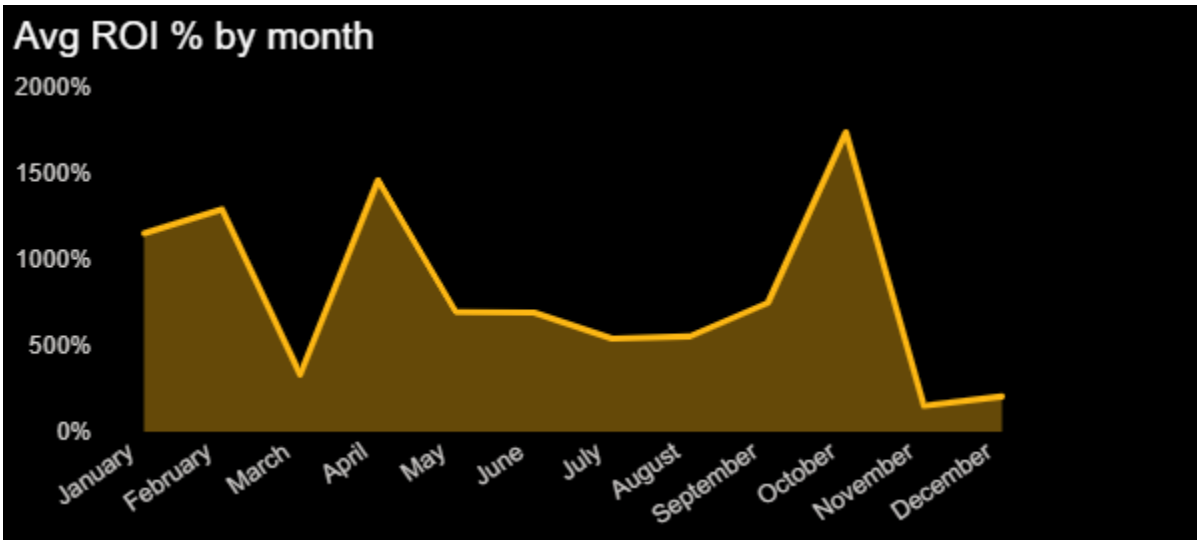


Standard feature-length runtimes outperform shorter films.

📅 4. Genre and Release Month

Release timing has a substantial impact on profitability.

Month	ROI
October	1,737%
April	1,460%
February	1,289%
January	1,151%

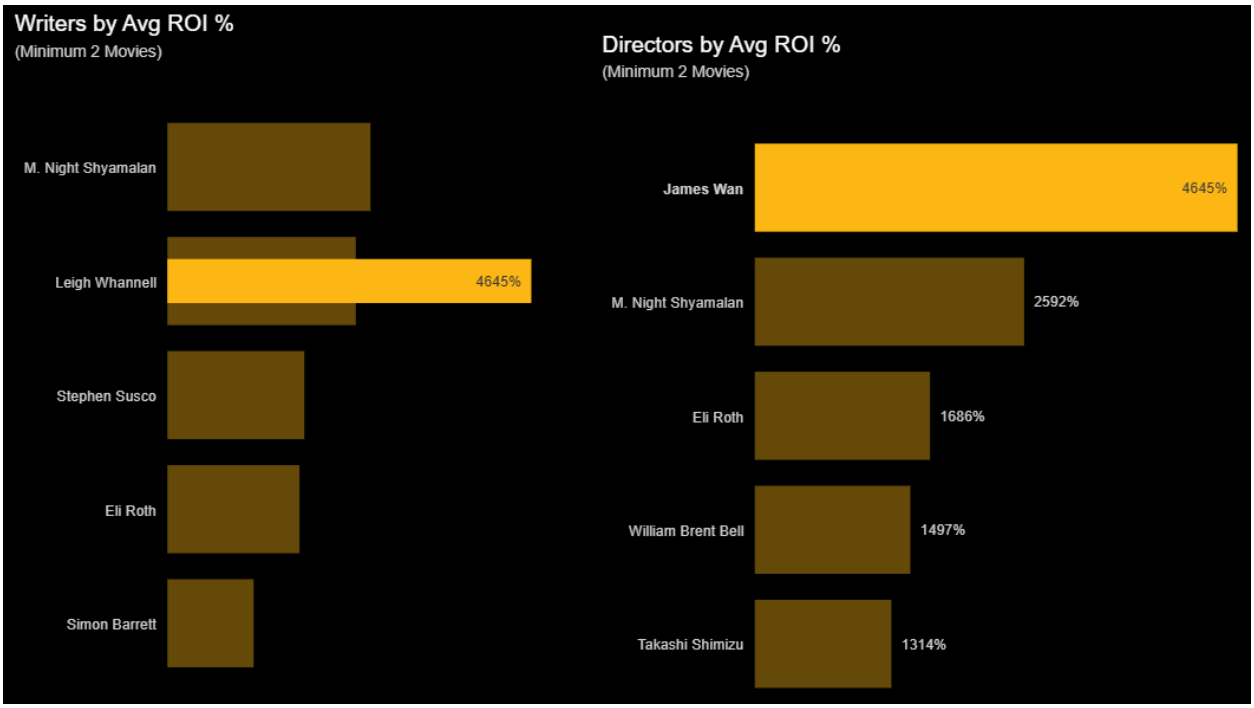


October clearly stands out as the strongest release window. April, February, and January are also very strong.

🎭 6. Writer and Director Impact

Creative leadership can dramatically amplify returns.

Talent	ROI
James Wan / Leigh Whannell collaborations	4,656%
James Wan	4,656%
Leigh Whannell	2,405%



Writer Leigh Whannell and director James Wan demonstrate consistently higher ROIs than others in their field, with their collaborations averaging **4,645% ROI**.

☑ Recommendations

• Focus exclusively on **low-budget horror films (≤ $10M)**
• Prioritize **PG-13 ratings**

• Target **90–120 minute runtimes**
• Release films during **October, January, February, or April**
• Seek experienced horror-focused talent such as **James Wan and Leigh Whannell**

🏁 Conclusion

By applying disciplined budget constraints and aligning production and release decisions with historically high-performing patterns, Fledgling Movie Studio can dramatically improve its chances of generating exceptional returns.

When these conditions align, ROI can **approach or exceed 1,000%**, far outperforming industry norms.

This analysis provides a **scalable, repeatable framework for building a profitable studio using data-driven decision-making**.